Radio and Mission Planner Setup

A clean setup sequence for the Radiomaster Pocket, ExpressLRS receiver, and Mission Planner validation workflow used to arm, mode-switch, and verify the final quad configuration.

Hardware and software used

Item	Purpose in the workflow
Radiomaster Pocket TX	Pilot input, mode switches, ELRS internal RF module, optional ELRS Backpack support
Happymodel EP2 RX	On-drone receiver for the ELRS control link
EdgeTX Companion	Create the multirotor model and assign channels / switches
ExpressLRS Configurator	Flash ELRS firmware for the transmitter and receiver
Mission Planner	Confirm radio bars, flight modes, arm/disarm channel behavior, and calibration

Radio firmware and binding workflow

- Use the top USB connector on the RadioMaster Pocket for computer interface; the bottom connector is for charging.
- Flash the internal ELRS module using EdgeTX passthrough in ExpressLRS Configurator. The final workflow used the RadioMaster 2.4 GHz target and kept the latest ELRS Lua script on the radio SD card under SCRIPTS/TOOLS.
- Apply the same binding phrase to both transmitter and receiver. The receiver can be updated over Wi-Fi after it powers on and enters hotspot mode.
- A good practice is to power on the transmitter first, then the drone, so the receiver binds quickly and predictably.

ELRS binding phrase workflow

- On the transmitter: run the ExpressLRS Lua script, enable Wi-Fi, connect to the radio hotspot, open 10.0.0.1, and set the binding phrase.
- On the receiver: power the receiver without a bound transmitter nearby, wait ~60 seconds for Wi-Fi mode, connect to the receiver hotspot, open 10.0.0.1, and set the same binding phrase.
- After saving and power-cycling, the receiver should show a solid LED when the radio link is established.

EdgeTX model configuration

- Create or duplicate a multirotor model in EdgeTX Companion and set the internal RF mode to CRSF for the standard control-link configuration.
- Use the standard channel ordering for a multirotor wizard profile: Roll = Ch1, Pitch = Ch2, Throttle = Ch3, Yaw = Ch4.
- Assign mode and arm/disarm functions to auxiliary channels. The final setup mapped flight modes to CH6 and arm/disarm to CH7, with a 3-position switch used for mode selection.
- Enable a switch warning on startup so the radio complains if powered on in an armed state. This helps catch unsafe switch positions before the drone is powered.

Suggested switch mapping pattern

Channel / switch	Suggested purpose	Notes
CH1-CH4	Roll / Pitch / Throttle / Yaw	Leave wizard defaults for the four main stick channels
CH5 / SA	Optional auxiliary feature	Can be used for a simple binary auxilary function
CH6 / SB	Flight mode select	Mission Planner can be configured to read flight modes from this channel
CH7 / SC	Arm / disarm	Set the RC7 function in ArduPilot to ArmDisarm and verify switch behavior carefully
CH8 / SD	Spare auxiliary	Reserve for later features or testing if needed

Mission Planner validation sequence

- Props off. Apply battery power to the flight controller so the receiver is fully powered, but remove propellers before any arm/disarm or motor-output checks.
- Open Mission Planner and confirm the radio is detected under Setup → Mandatory Hardware → Radio. Move both sticks and verify that the Pitch, Yaw, Roll, and Throttle bars respond correctly.
- Confirm FLTMODE_CH = 6 so the intended switch changes the vehicle flight modes. Then open the Flight Modes page and move the assigned switch to verify the mode transitions.
- Configure RC7_FUNCTION = ArmDisarm and check that the assigned arm/disarm switch produces the expected behavior and spoken notifications.
- Run the radio calibration page if needed: click Calibrate Radio, move the sticks through full travel, and save the calibrated values.

Mission Planner quick checks

Use the Mission Planner data view to confirm that the mode text changes when the mode switch is moved. Enable speech for arm/disarm and mode announcements on the Planner settings page, it makes initial bench checks much easier.

Mission Planner parameter quick reference

Parameter	Typical value used in the build	Why it matters
FLTMODE_CH	6	Maps the selected auxiliary channel to flight mode selection
RC7_FUNCTION	ArmDisarm	Makes the chosen switch toggle arm/disarm logic
BRD_ALT_CONFIG	1	Needed for the board-specific ELRS RX2/TX2 remap
RC_PROTOCOLS	1	Allows automatic RC protocol detection
SERIAL6_BAUD	460	Used for ELRS-based serial link configuration
SERIAL6_PROTOCOL	23 (CRSF) or 2 (MAVLink)	Determines whether the ELRS receiver carries standard CRSF control or MAVLink telemetry mode

Controller evidence images

Front view of the configured Radio Master Pocket transmitter used with the final build.

Top-port view highlighting the USB interface used for radio firmware and configuration workflows.